

Lead Prioritization Agent

Introduction

This demo is a simple concept design for scoring leads in B2B cyber solutions. It is from the perspective of a sales operations team member.

The demo supports account executives (AEs) and regional directors (RDs) by leveraging modeling and agentic AI to score and prioritize sales leads. It produces a lead prioritization score from 1-100 for open leads, using temporal-based textual representations and in-context comparisons for LLM reasoning and knowledge. It includes explanations of scores as well as recommended steps to address the lead opportunity.

Account Prioritization

We define **account priority** by the amount of attention a lead requires, based on its propensity to convert and deliver revenue.

For this demo, we define our evaluation of account prioritization to be based on 3 components.

1. Customer Profile
2. Engagement and Shopping Behavior
3. Pipeline Stage

Customer Profile

We determine basic firmographics, such as industry, scale, core business, and market position, which can be highly predictive of risk and cybersecurity need. Scale can be proxied by revenue or employee count. We also consider the lead's digital maturity, technology investment, and specific technology stack, which can also indicate fit. Company website, news, and filings may reveal this information. Another key component may be past threats and breaches, which would indicate awareness. Customer profile ultimately determines if our company can fulfill their need, and if the company can afford and deliver revenue to our sales pipeline. Customer profile is generally less dynamic, and does not change in the short-term.

Engagement and Shopping Behavior

To track buying intent, we want to track whether engagement is direct or indirect, the quality of engagement, and engagement trends. Conference attendance and website visits to our demos and case studies can signal an interest, while direct interactions may provide an even stronger indication.

Additionally, less structured tracking, such as social media activity and job posting activity may also indicate a current or future need.

As part of tracking engagement, we also want to track the level of decision-making authority in the engagements. Typically, a higher seniority indicates increased attention and intent.

Finally, we also track lead competitor businesses. Understanding their current solutions, renewal dates, pricing, and RFP activity can indicate opportunity.

Pipeline Progress

For this demo, we define the lead pipeline stages as: Prospecting, Qualified, Proposal, Negotiation, Conversion. Leads lost are in stage: Lost. Typically, the further along an open lead is in the pipeline, the greater its propensity to convert and deliver revenue. However, a lead that has stagnated may indicate a lower propensity.

Assumptions

Assumptions for this demo.

1. Focuses only on new business, no renewal business.
2. Does not differentiate between our offerings.
3. Uses select core features of customer profile, engagement, and pipeline progress for evaluation.
4. Conversion for 1 year revenue. No explicit consideration of retention value.
5. Website visits are sales-oriented. Interactions are conference, email, or calls interactions.
6. No significant temporal distributional shift in leads over years. That is, the statistical properties of lead attributes (e.g., engagement, firmographics, behavior) are stable over time.
7. Present day is 2025, May 1

Data and Methods

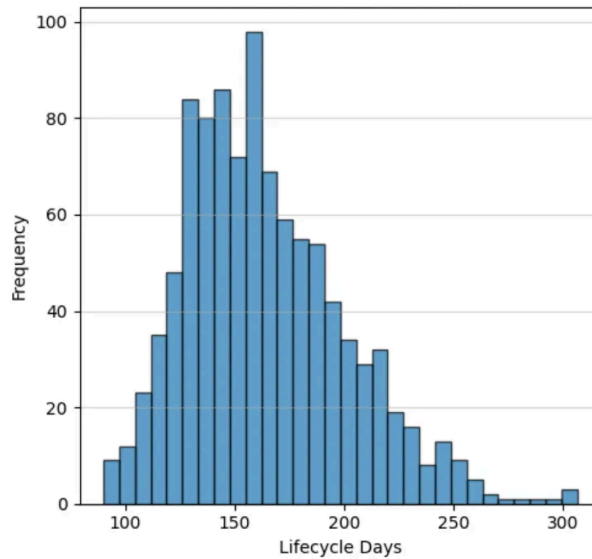
Synthetic and real data is used for this demo. The company names and industry titles are real, and were generated from the following sources:

1. Known customers listed on the company website (55 records)
2. GPT-generated customer competitors (101 records)
3. Beach, J. (2023). *Top 2000 Companies Globally* [Data set]. Kaggle. (250 records)

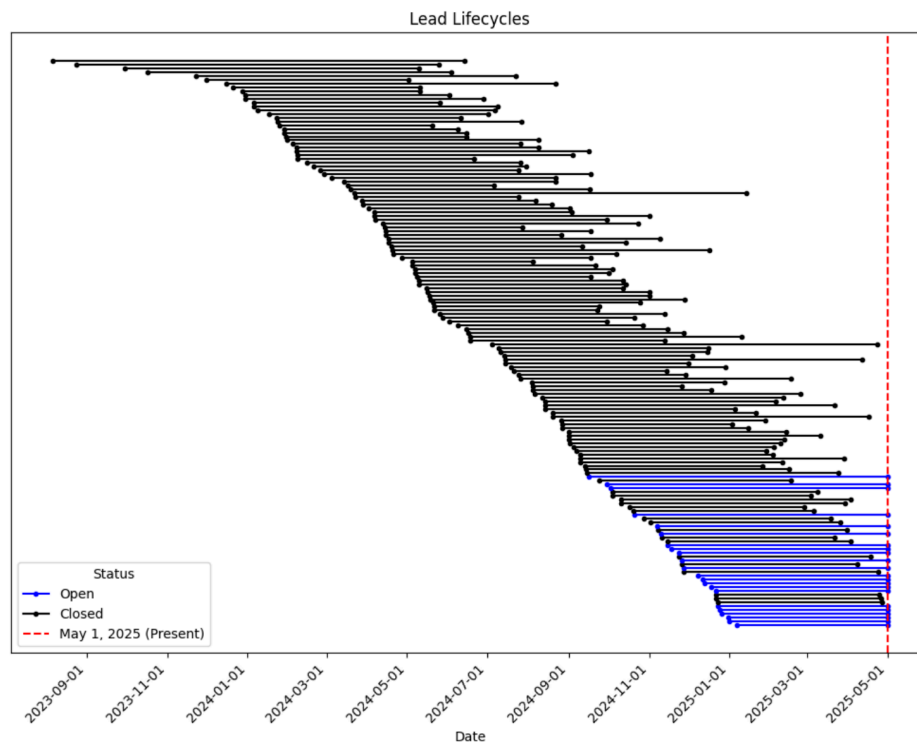
Customer competitors were added for comparability between leads, and under the assumption that these would match the customer profile well. Customers are global, spanning across industries such as Finance, Technology, Healthcare, Education etc., and ranging from middle market to major enterprises.

Simulating Leads

To simulate real lead data, we first assume a present date of May 1, 2025. Each account is given a random opening date within the past 2 years, with a lead lifecycle spanning between 3-12 months — skewed to the right.



Leads are open if they are open as of present May 1. (Shown in blue lines below)



Application Architecture and Process Flow

At baseline, we can leverage modern chatbot style LLMs to evaluate sales leads based on their pretrained sales and industry knowledge. This can be done by inputting the subject open lead's data with a prompt for a priority score, explanations, and recommendations.

However, the baseline approach has several drawbacks. First, this does not leverage the expertise in our own customer profile, nor does it compare the subject leads of interest to our own historical lead patterns. Furthermore, high-level summaries lack the nuance of trend, which can be a major predictor in lead movement.

Leveraging Historical Comparables

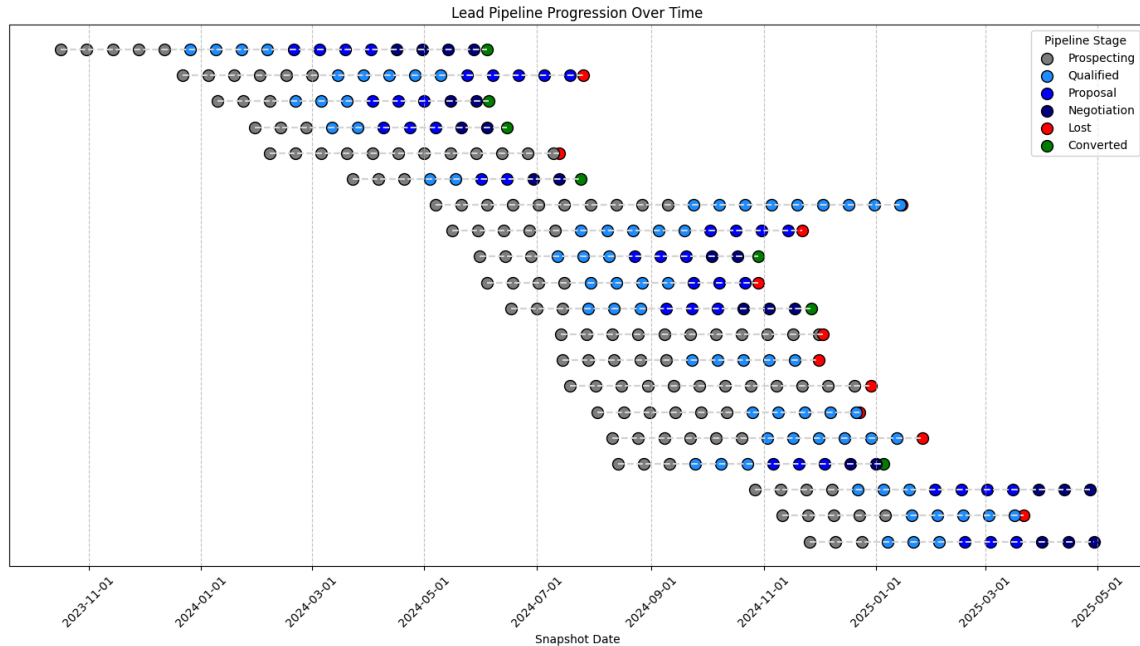
For this demo, we will consider a lead a historical comparable if it has closed (Converted or Lost). By comparing an open lead to similar historical comparables, we can develop a data-backed scoring algorithm. Because sales leads are complex and high variable, and guided by sets of rules known well by experts, agentic language models such as GPT are well suited to comparing leads against historical profiles with quality reasoning.

This can be achieved by a few-shot learning technique called in-context learning. In this method, we send an open lead along with top k most similar historical comparables with their outcomes, so the AI can learn from the examples to evaluate the open lead.

Before this in-context learning evaluation can be achieved, we first design an algorithm to identify an open lead's top k comparables. We utilize OpenAI's *GPT-3.5-turbo* to first summarize every lead in our database, accounting for company profile, trends and patterns, and other aforementioned features.

Capturing Trends

To ensure our model captures trends, we can capture regular "snapshots" of leads during their active phase, at regular intervals and at milestone events. For demo purposes, we simply capture snapshots at 2-week intervals, tracking new developments on engagement, contact seniority, estimated deal value, competitor solutions, renewal dates, and pipeline advancements.

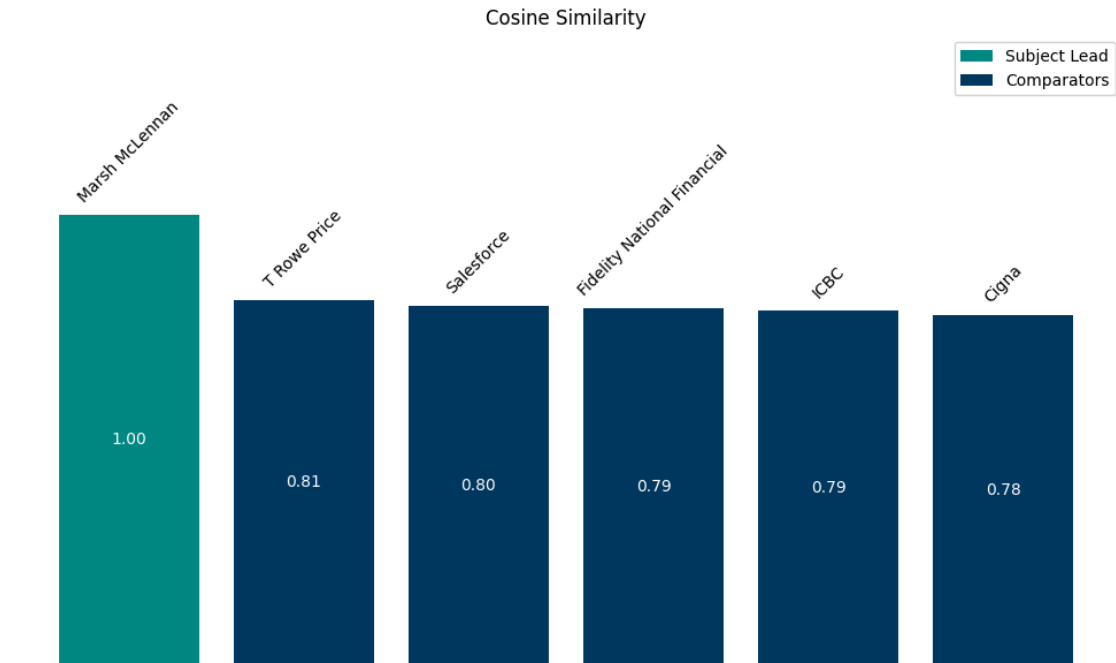


In this figure (above), each marker is a snapshot, taking place every 2 weeks.

This overcomes a major limitation in generating LLM summaries from only the most recent state of leads. For in-context learning, more snapshots provides a much richer field of data points to draw historical comparables from. For example, an early qualified lead for a middle-size healthcare company can be compared to other healthcare companies in their qualified stages. Hence, the lead summaries are generated at every single snapshot, accounting for the full lead histories prior to it for each summary.

Similarity Matching Snapshots

With snapshot summaries generated, we embed the summaries using a lightweight but powerful semantic embedding model, *all-MiniLM-L6-v2*. We then utilize cosine similarity between the open lead and all historical comparables embeddings to find the most similar $k=5$ leads, ordered from most to least similar, using a similarity threshold of 0.8. Each lead ID can only be selected once in the matching pool.



Agentic AI Priority Scoring

For the priority scoring agent, we utilize the *GPT-4* model, due to the increased need for increased memory, creativity, complexity, and accuracy.

We provide the AI with the subject lead, and include the 5 comparables in order of similarity, along with the the conversion outcome of the comparables. We prompt the model to evaluate the quality of a sales lead to provide a prioritization score, using both historical comparables and broader sales knowledge. For each lead prioritization component, profile, engagement, and pipeline, the AI is prompted to provide a score from 0-100, along with an explanation

Finally, it is prompted to using its evaluation as a guide, provide a priority score from 0-100 to "reflect how much attention this lead deserves." As part of this evaluation, the AI is also prompted to provide some recommendations to improve the success of the lead.

Weighted Scoring

An alternative to the final periodization score from *GPT-4* is weighted average scoring of the components. This would allow AEs and RDs to exert greater overall control of the scoring rules, and can be adapted to shifting company goals or even just due to experience and heuristics. This approach can eliminate a source of variability, and provide users with a more granular understanding of lead prioritization metrics.

Neural Networks

While *GPT-4* may be effective at in-context learning, reasoning and explainability, its inherent design for randomness and language prediction makes it inconsistent at scoring. While its evaluation predicts a consistent sentiment for lead evaluation, the numerical aspect of language models is not "faithful" to

math. While no sensitivity analysis has been conducted, we can hypothesize that data sparsity issues in historical comparables and summarization biases can have significant influence on the evaluation, especially on the numerical outputs.

A potential supplement is to leverage a neural network model, such as a deep-learning model that can infer conversion or prioritization based on snapshot summaries. Another sophisticated approach could leverage transformers and long short-term memory (LSTM) models to track trends. These modeling approaches can be trained on snapshot history patterns as well as overall market and portfolio trends with a time-series training approach. While these supplemental approaches would be far less explainable, they would provide more consistent, “math-faithful” outputs.

Essentially, here are 3 approaches.

1. Prompt *GPT-4* for a priority score.
 - Highly explainable and already built into process flow, but lacks mathematical substantiation.
2. Calculate a priority score based on *GPT-4*’s individual component scores.
 - Narrows the source of variability, and is based on a clear SME understanding of component importance and weight. If heuristics change slightly, the scores can be recalculated.
3. Deep Learning Model
 - Can be an effective supplement to the language model output. Higher complexity than language reasoning, and potential for great performance. However, it lacks explainability which can be crucial for AEs who must interact with lead contacts.

Implementation

In the initial evaluation of the model, AEs and RDs can test the correlation between the model outputs against existing lead metrics and evaluate the rationalizations of the model. This feedback can be used to form better snapshot summaries, emphasizing information that is more predictive and relevant, and avoiding information that is not. It can also guide the prompting of the final prioritization evaluations to increase consistency and reasoning accuracy.

One major concern, however, is that the model is highly iterative on engagement and AE actions, which can blur the understanding of model performance. There is a risk that if the model is relied on early, it may influence AEs to cause particular leads to convert at higher rates despite truly better leads elsewhere. Concurrently, this may cause quality leads to be missed when the model did not prioritize them. This may eventually cause self-fulfilling prophecy where the model appears to be performing strongly, when it is actually not optimizing sales workflows.

Recommendations

Background Implementation

Implement the model silently, without users having access to results that can influence their decision-making. Users can write occasional snapshot summaries or write reports that can be compared to the model output. Over time, conversion and lead value can be compared against model recommendations to determine matches and missed opportunities.

This method would have delayed benefit, but prioritizes understanding of the model strengths and weaknesses for improvement.

A/B Testing

Implement the model to some AEs. Randomize access based on AE portfolio similarity, so that performance can be compared.

While this would immediately influence real-world decisions, it may be unrealistic to gather enough AEs for enough time for there to be statistical power in determining statistical significance. Furthermore, if incentives and performance are tied to the model, it may be unfair to implement.

Blinded Recommendations

If AEs are unaware of the model, providing them with suggestions or nudges towards model-recommended accounts can be used to observe if AEs naturally gravitate towards these leads. If they do, the model may be identifying opportunities. However, if they do not, it could also indicate that AEs are not looking to change their current prioritization system.

Full Transparency

Communication on the AE's reports of model performance and influence on their prioritization decisions. Reflective analysis on blind spots, strengths, and weaknesses of the current systems and new model implementation.

Conclusion

Summary of Process Flow

Background

1. Around 400+ companies generated, with closed and open leads spanning ~2 years.
2. All leads are snapshotted at 2 week intervals until closed
3. Structured text summaries are generated with *GPT-3.5-turbo* for each lead snapshot
4. All summaries are vectorized by embedding model *all-MiniLM-L6-v2*. Closed leads are historical comparables.

Evaluating Subject Lead

1. Subject lead (open lead of interest) is snapshotted and summarized
2. Summary is vectorized by embedding model *all-MiniLM-L6-v2*
3. Cosine similarity calculates the most similar $k=5$ snapshot summaries to subject lead summary.
4. Subject lead summary as well as comparable summaries and outcomes are input to *GPT-4*, which uses comparables for in-context learning. The AI then produces quality scores for customer fit, engagement quality, and pipeline progress using comparables and broader sales knowledge. Finally, the AI provides a priority score as well as recommendations.

Future Steps

5. Neural network evaluates historical snapshot summaries to infer a priority score of the subject lead, leveraging the entire span of historical context and time-series learning. This would supplement the language model.

Future Considerations

Data Quality

Currently the summaries are based on basic instructions to *GPT-3.5-turbo*, without any fine-tuning examples of effective summaries for our specific business model. Future summaries may be annotated or fully structured and written by internal sales experts to improve relevance and comparable identification.

A future agent feature for summaries and lead data in general is updating the data based on discussion and correction from AEs and other users. The sales experts often have ground truth understanding of leads, and the AI agent can iterate on their knowledge to constantly check and improve data quality.

The cosine similarity matching algorithm is relatively simplistic, using a basic total embedding and matching. Future embedding-matching algorithms can be more specific, capturing specific features that hold greater comparable relevance. For example, if industry is a much better reference for how a lead should be evaluated, a weighting or clustering algorithm can account for this. Techniques include separately weighted embeddings for different sections of text and unsupervised k-means clustering.

Deeper Context in Learning Comparables

Currently, the final evaluator step of the model uses in-context examples which are just snapshot summaries of the most similar comparable snapshots. Future models that can handle more text may include the future state of the comparables, indicating not just whether the comparable converted or not, but what stage it advanced to and why it may have converted or lost.

Handling Temporal Trends

Market conditions and trends, as well as internal factors such as ICPs, strategies, and products change over time. In the long-term, time between subject leads and historical comparables may increasingly lead to bias, driving the need for temporal distance in similarity calculations. Capturing trends and

allowing the AI agent to learn the trends may also maintain the model's ability to accurately predict conversion success and infer lead prioritization.

Application Agent Demo

Start Here

Search a Lead

Clear All

Enter the company name or lead ID (last 4 digits):

0017

Company: Pendo

ID: ACC0017

Status: Open

Open Date: 2024-12-16

Close Date: None

Lead Stage: Negotiation

Summary: Company Summary: Pendo is a technology company that specializes in offering a Digital Adoption Platform to help businesses enhance user experiences and maximize software utilization. With over 500 dedicated employees worldwide and annual revenues exceeding \$100 million, Pendo has firmly established itself as a leader in the industry. As a rapidly growing company, Pendo is continuously innovating and expanding its market presence.

In terms of technology, Pendo is speculated to have a sophisticated data infrastructure and high digital maturity, allowing for advanced analytics and insights into user behavior. The company likely leverages cloud platforms extensively to scale its services and ensure seamless integration with various software applications. Overall, Pendo's commitment to cutting-edge technology and data-driven solutions positions it as a frontrunner in the competitive digital adoption landscape.

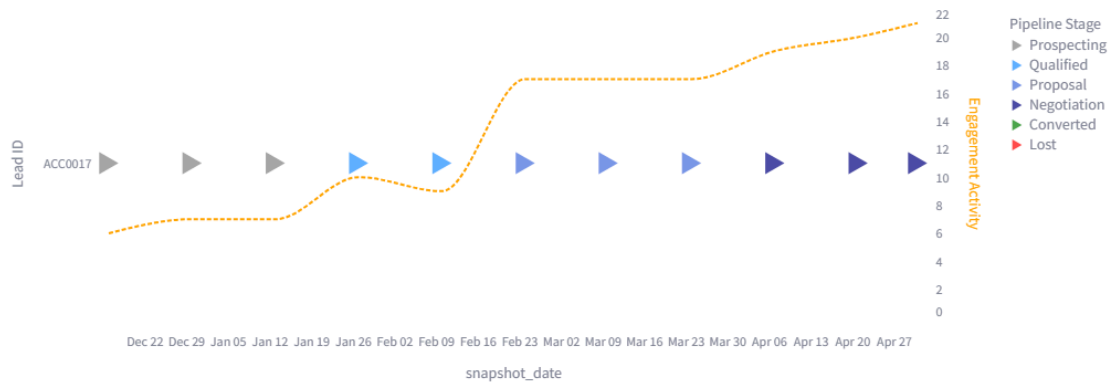
Search a Lead

Engagement:

In terms of engagement, downloads remained consistent at 3, website visits increased from 3 to 7, and interactions increased from 2 to 11. There was a notable spike in interactions during the Negotiation stage.

Senior Contact:

The current most senior contact holds the title of Exec/CIO/CISO. There was a recent progression in the seniority of the most senior contact 4 weeks ago when the contact changed from a Director to an Exec/CIO/CISO during the Proposal stage.

Lead Stage Progression and Engagement Over Time

Lead Timeline and Engagement

Ask about this lead

Score Lead

Ask a question about this lead:

What next steps do you recommend?

Question: What next steps do you recommend?

Answer: Based on the information provided and the progress of lead contention, here are a few next steps I recommend:

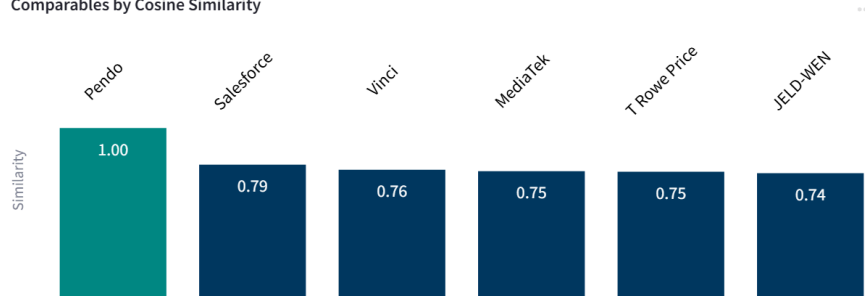
1. **Finalize Negotiations:** As the lead has been in negotiation stage for 3 weeks, efforts should be made to finalize the deal terms and conditions. This can be achieved by addressing any outstanding concerns or blockers that the lead may have.
2. **Continue Engagement:** The lead has shown consistent engagement with downloads, website visits, and interactions all increasing over time. Continue this engagement by providing more content that could be useful for them. New product updates, webinars, case studies, or any industry-related content might help keep the lead engaged.
3. **Leverage the Decision Maker:** It's essential to leverage the decision-maker's involvement in this lead. As the most senior contact is an Exec/CIO/CISO, ensure that any communications are appropriately targeted, and that the value proposition is clear to them.
4. **Prepare for Competitor Renewal Date:** With competitor's renewal date coming in the next five months, focus on pulling forward our proposed solution's value and differentiate it from the competitor's offering. We should aim to close before the renewal comes into play.
5. **Offer a Demo or Trial:** If it hasn't been offered already, providing a free demo or trial of your services could be an effective method to show the value and efficiency of your solution.

Agent Questions

Question: Does this lead have a competitor solution?

Answer: Yes, this lead has a competitor solution. However, the specific details of the competitor solution are not provided.

Comparables by Cosine Similarity



Scoring Lead:

Ratings:

1. Company Fit: 85

Pendo aligns well with the company's ideal customer profile. Being a technology company with high digital maturity and a strong focus on data, Pendo's technological profile aligns with our offering. Their high cyber investment level, deal value, and size compare favorably with successful comparable leads. However, the lead's deal value is slightly lower than the average deal value of the comparable converted leads, which limits the score.

2. Engagement Quality: 90

Pendo has shown a steady increase in engagement (website visits, interactions) over time, especially during the Negotiation stage. The seniority level of the contact has also progressed to a high level (Exec/CIO/CISO), which is a strong indicator of serious interest. The knowledge of a competitor solution and its renewal date provides an opportunity for strategic engagement. The score reflects these positive aspects.

3. Pipeline Progress Potential: 85

The lead is currently in the Negotiation stage, which is late in the sales pipeline, indicating that it is close to conversion and deserves attention. However, the lead has been in this stage for 3 weeks, which is slightly longer than most comparable leads that converted. Despite this, the lead's progress mimics the pattern of comparable leads that converted, offering a high potential for conversion.

Overall Lead Priority Score: 87

This lead shows a high level of engagement, aligns well with the company's customer profile, and is in a late stage of the sales pipeline. Therefore, it deserves high attention from the sales team.

Cosine Similarity and Priority Score

References

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